Raghav Singh

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Education

Indian Institute of Information Technology, Nagpur — Nagpur, India

Bachelor of Technology in Data Science and Analytics — CGPA: 5.04/10 — *Aug 2023 – May 2027*

- **Coursework:** Database Management Systems, Operating Systems, Computer Networks, Cloud Computing, Data analyst, Computer Architecture

Technical Skills

- **Programming Languages:** Python, JavaScript, TypeScript, C++, SQL, Bash
- **Web Technologies:** React.js, Flask, FastAPI, HTML, CSS, Node.js
- **Machine Learning & AI:** PyTorch, Scikit-learn, OpenCV, Computer Vision, Deep Learning, CNN
- **Cloud & DevOps:** AWS (EC2, ECS, Fargate, S3), Docker, Google Kubernetes Engine (GKE), GitHub Actions, CI/CD
- **Databases:** PostgreSQL, MongoDB, SQLite, Firebase
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Streamlit
- **Tools & Technologies:** Git, VS Code, Postman, NetworkX, RESTful APIs, Reverse Proxy
- **Soft Skills:** Problem-Solving, Analytical Thinking, Team Collaboration, Communication, Active Listening

Projects

Hierarchical Deep Learning for Drug Response Prediction in LUAD — *5th Sem – 2025*

Technologies: Python, PyTorch, Transformers, ChemBERTa, SHAP, PCA, GDSC

- Curated 3,190 LUAD drug-cell line samples from GDSC database and performed gene expression preprocessing with PCA dimensionality reduction (5 PCs, 96.6% variance)
- Engineered multimodal features combining PCA-compressed gene data with 768-dimensional ChemBERTa embeddings using transformer-based cross-attention
- Built hierarchical classification-regression pipeline with focal loss, achieving AUC 0.906, accuracy 0.875, and R^2 of 0.291 on test set

Deployr – Full Stack Hosting Platform — *Jan 2025*

Technologies: Docker, AWS ECS, AWS Fargate, Amazon S3, Google Kubernetes Engine, Reverse Proxy, GitHub API

- Developed a comprehensive full-stack web hosting platform supporting static and dynamic applications with GitHub integration and real-time configuration management
- Architected scalable cloud infrastructure using Docker, AWS ECS/Fargate, S3, and Google Kubernetes Engine for orchestration
- Implemented reverse proxy routing and CI/CD via GitHub webhooks, achieving 99.9% uptime and reducing deployment time by 60%

Pothole Detection System — *Jan 2025*

Technologies: PyTorch, OpenCV, CNNs, Flask, React.js, Computer Vision

- Built a deep learning-based computer vision system for automated pothole detection using Convolutional Neural Networks
- Achieved real-time inference with 92% accuracy using optimized CNN models deployed via Flask REST API
- Developed an interactive React.js dashboard for visualization and real-time monitoring

Delivery Route Optimization System — *Dec 2024*

Technologies: Python, NetworkX, Graph Algorithms, Dijkstra, Bellman-Ford

- Built an intelligent delivery optimization engine implementing Dijkstra's and Bellman-Ford algorithms for shortest and cost-efficient routes
- Integrated real-world map data using NetworkX, improving route efficiency by 35% and reducing delivery time by 25 minutes per route
- Designed scalable architecture handling 1000+ nodes and 5000+ edges with optimized computation time

Global Renewable Energy Data Analysis (1965–2023) — *Nov 2024*

Technologies: Python, Pandas, NumPy, Power BI, Streamlit, Statistical Analysis

- Analyzed 50+ years of renewable energy data across 150+ countries to identify adoption trends in solar, wind, and hydroelectric power
- Processed 500,000+ data points using Pandas and NumPy, revealing 300% growth in renewable capacity
- Built interactive Power BI dashboards and Streamlit web applications serving 1000+ monthly users for public data exploration

Open Source Contributions

- Active contributor to open-source projects on GitHub with experience in collaborative development, code reviews, and issue resolution
- Contributed to backend services optimization, workflow automation, and comprehensive documentation improvements
- Participated in pull requests following best practices for version control and agile development methodologies